3.1.1 Atomic structure: Sheet 2

Isotopes and mass spectrometry · 19 marks · about 25 minutes

Key information

- **A_r** = Σ(m/z × abundance) ÷ total abundance, compared with 1/12 of the mass of a ^{12}C atom.
- Electron impact: $X(g) + e^- \rightarrow X^+(g) + 2e^-$. Electrospray: $X(g) + H^+ \rightarrow XH^+(g)$, so subtract 1 from the m/z to get M_r.
- All ions get the same kinetic energy, so lighter ions travel faster and reach the detector first.
- **KE = ½mv²** and **v = d ÷ t**. Mass of one ion in kg = (mass of 1 mol in g ÷ 1000) ÷ L.
- Ions gain an electron at the detector, creating a current; abundance ∝ size of the current.

Section A: Self-quiz

01 A sample of bromine (Br_2) is ionised using electron impact ionisation. Give an equation, including state symbols, for the process that occurs during the ionisation of bromine.

[1 mark]

02 State how the detector enables the relative abundance of each ion to be determined.

[1 mark]

03 Which statement about time of flight mass spectrometry is correct?

[1 mark]

A The current in the detector is proportional to the ion abundance
B Sample particles gain electrons to form positive ions
C Particles are detected in the order of their kinetic energies
D Ions are accelerated by a magnetic field

04 Samples of four different substances are analysed using time of flight mass spectrometry. In each case, the samples are ionised to form ions with a single positive charge. The ions are accelerated to the same kinetic energy. Which sample gives ions with the shortest time of flight?

[1 mark]

A A sample of ^{45}Sc that is ionised using electron impact ionisation.
B A sample of C_3H_8 that is ionised using electrospray ionisation.
C A sample of CH_3CH_2OH that is ionised using electrospray ionisation.
D A sample of CO_2 that is ionised using electron impact ionisation.

05 A mixture of compounds with molecular formulas $C_3H_5O_2N$, $C_3H_7O_3N$ and $C_3H_7O_2NS$ is ionised using electrospray ionisation.

Give the formula of the ion that reaches the detector first in the TOF mass spectrometer.

[1 mark]

Section B: Exam questions

06 A sample of tin is analysed in a time of flight mass spectrometer. The sample is ionised by electron impact to form 1+ ions.

Table 1 shows data about the four peaks in this spectrum.

Table 1

m/z	Percentage abundance
112	22.41
114	11.78
117	34.97

m/z	Percentage abundance
120	To be determined

Give the symbol, including mass number, of the ion that reaches the detector first.

Calculate the relative atomic mass of tin in this sample.

Give your answer to 1 decimal place.

[4 marks]

07 Compound X is dissolved in a polar, volatile solvent and is ionised by electrospray ionisation.

Each ion is accelerated so that it has a kinetic energy of 1.36×10^{-16} J

The kinetic energy of an ion is given by the equation $KE = \frac{1}{2}mv^2$ (KE / J, m / kg, v / m s^{-1})

The time of flight along the 0.750 m flight tube is 2.48×10^{-5} s

Determine the mass, in g, of one mole of X.

The Avogadro constant, $L = 6.022 \times 10^{23}$ mol^{-1}

[5 marks]

08 A mixture of argon, carbon dioxide, nitrogen and oxygen is analysed using TOF mass spectrometry. The mixture is ionised by electron impact.

Identify the ion formed from this mixture that reaches the detector last.

Justify your answer.

[2 marks]

09 A sample of nickel was analysed in a TOF mass spectrometer. The sample was ionised by electron impact ionisation. The spectrum produced showed three peaks with abundances as set out in Table 2.

Table 2

m/z	Abundance / %
58	61.0
60	29.1
61	9.9

Give the symbol, including mass number, of the ion that would reach the detector first in the sample.

Calculate the relative atomic mass of the nickel in the sample.

Give your answer to one decimal place.

[3 marks]

Mark Scheme

M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

Question	Marking guidance	Comments	Mark
01	$Br_2(g) \rightarrow Br_2^+(g) + e^-$ OR $Br_2(g) + e^- \rightarrow Br_2^+(g) + 2e^-$	Source: 7404/1 Jun 2024 Q06.2 Do not penalise the inclusion of a radical dot	1
02	The (relative) abundance is proportional to the size of the current	Source: 7404/1 Jun 2024 Q06.4	1
03	**A**	Source: 7404/1 Jun 2017 Q12	1
04	**D**	Source: 7404/1 Jun 2024 Q20	1
05	$C_3H_6O_2N^+$ / $C_3H_5O_2NH^+$	Source: 7404/1 Jun 2019 Q02.2 Must be charged	1
06	M1 $^{112}Sn^+$ M2 missing abundance = 30.84% M3 RAM = [(112 × 22.41) + (114 × 11.78) + (117 × 34.97) + (120 × 30.84)] ÷ 100 M4 RAM = 116.5	Source: 7404/1 Nov 2021 Q01.2 If M2 missing then allow M3 if denominator = 69.16 Allow M4 ECF; answer must be to 1 dp	4
07	M1 $v = d \div t = 0.750 \div 2.48 \times 10^{-5} = 30\ 241.9\ m\ s^{-1}$ M2 $m = 2KE \div v^2 = (2 \times 1.36 \times 10^{-16}) \div 30\ 241.9^2 = 2.974 \times 10^{-25}$ kg M3 $m = M2 \times 1000 = 2.974 \times 10^{-22}$ g M4 mass of one mole of ions = M3 × 6.022×10^{23} = 179(.1) M5 subtracts 1 for mass of H^+: mass of one mole of X = 179.1 – 1 = 178(.1) g	Source: 7404/1 Jun 2023 Q07.1 Allow ECF at each stage	5
08	M1 CO_2^+ M2 Has the highest mass (to charge ratio) (so will travel the slowest)	Source: 7404/1 Jun 2023 Q07.3	2
09	M1 $^{58}Ni^+$ M2 $A_r = [(58 \times 61.0) + (60 \times 29.1) + (61 \times 9.9)] \div 100$ M3 $A_r = 58.9$	Source: 7404/1 Jun 2017 Q01.4 M1 needs mass and charge M3 must be to 1 dp	3